

lec58

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1 Review of 2-Sample Hypothesis Test, Efficiency, Robustness

- Recall that given $\bar{X}_C$ and $\bar{X}_T$, the two-sample test for no difference in means is given by

$$z = \frac{(\bar{X}_T - \bar{X}_C) - 0}{\frac{\sigma_T}{\sqrt{n_T}} + \frac{\sigma_C}{\sqrt{n_C}}}$$

Then, we compare

$$2 \cdot P(Z > z) < \alpha$$

- Efficiency** trusts the data, and tries to use as much data that the original setup provides as possible to reach a conclusion.
- Robustness** protects against bad assumptions, noise, or inaccurate data.

2 Efficiency and Robustness with Regularizers

2.1 Ridge Regression

Consider the objective of ridge regression:

$$\arg \min_w \sum_{i=1}^n |y - w^T x|_2^2 + \lambda |w|_2^2$$

This is equivalent to putting a **Gaussian prior** on the weights, and assuming

$$w \sim N(0, \frac{1}{\lambda} I)$$

which results in the optimal

$$\hat{w} = (X^T X + \lambda I)^{-1} X^T y$$

Ridge regression penalizes large w_i s with higher loss.

- Therefore, a larger λ means that we trust the prior more, and thus shrink w_i s to 0 $\implies$ robustness
- Conversely, a smaller λ means that we trust the prior less, and thus allow for greater w_i s $\implies$ efficiency

2.2 LASSO Regression

- LASSO regression also introduces penalties for large weights.
- However, rather than shrinking weights, LASSO regression pushes coefficients exactly to 0, and thus keeps contributions only from important weights.

The objective of LASSO regression is the following:

$$\arg \min_w \sum_{i=1}^n |y - w^T x|_2^2 + \lambda |w|_1$$

where

$$|w|_1 = \sum_{i=1}^k w_i$$

The same concepts of efficiency and regularization also pertain here. A larger lambda means we trust LASSO prior more, and shrink more coefficients to 0. A smaller lambda means that we trust the LASSO prior less, and keep more coefficients.

3 Distributional Robustness

- Recall that a regression model is given by

$$\hat{y} = \hat{m}X + \epsilon$$

- However, what if there is a small subset of data that are adversarial to our model, i.e. they lead our data to be inaccurate.
- Solution 1: Acquire more data to increase robustness
- Solution 2: Reduce the influence of one corrupted data point by converting to percentiles or ranks
 - Reduces efficiency but increases robustness

4 Three Types of Priors

- **Smoothing prior** - Assume that w_i s are close to 0 - use regularization
- **Corruption prior** - Assume that some data points are corrupted or extreme, and can thus affect the final accuracy of our predictions - convert to percentiles, ranks
- **Low-Rank prior** - Assume that the actual variation only occurs in a small subset of the original predictors - use dimensionality reduction methods such as PCA
 - In PCA, each principal component (PC_i) captures the i th direction with the most variance in the dataset.